

LAB 2: Astronomy with the 21-cm Line and a Horn Antenna

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1. Introduction

This lab extends the signal processing techniques from Lab 1 to astronomical measurements using a horn antenna. We will tune to frequencies around 1.4 GHz to measure 21-cm hyperfine emission from neutral hydrogen in the Milky Way. With luck, we will see evidence of the spiral-arm structure of gas in our galaxy.

In the second half, we will study signal propagation through transmission lines. We will characterize the cables and reflections in our receiver system to calibrate our measurements.

Organize your work to produce a high-quality L^AT_EX report addressing the goals in §2. Succinctly describe your experimental setup and hardware. Include new sections describing your observations, calibration, analysis, and statistical error estimation.

Suggestions for organizing your work:

- Develop version-controlled command-line scripts for data acquisition. Save data files with **all necessary metadata** to reconstruct observations. Err on the side of recording too much to avoid re-observing.
- Test your reality. Verify that rotation matrices yield correct pointings and that gain calibration is accurate. Build confidence through internal consistency checks.
- Think critically about assumptions and approximations. How do pointing accuracy and aperture coverage affect your calibration? Critically analyze these and other factors in your report.
- Report empirical results with error bars. (Values without confidence intervals are meaningless.) Estimate accuracy by comparing your model to the data. Quantify systematic and statistical sources of error. Would you bet money on your reported accuracy? If not, what sources of error (systematic or statistical) are undermining your confidence?

2. Goals and Instructions for Your Report

Lab reports and analysis code must be written *individually*. Include a block diagram of the telescope receiver in your report. You may draw this by hand and scan it, or use (old-school) software like `xfig` or another drawing tool.

Your report should demonstrate mastery of these learning goals:

- **Time:** Accurate conversion between UTC, PST, LST, and Julian Day.
- **Pointing:** Use rotation matrices to convert among spherical coordinate systems.
- **Measurement:** Measure the 21-cm line power spectrum from Milky Way hydrogen at a defined location.
- **Calibration:** Calibrate observations to an absolute scale, remove systematic instrumental effects, and apply noise-reduction techniques.
- **Doppler Correction:** Produce spectra with velocities calibrated relative to a standard reference frame.
- **Fitting:** Fit spectra with Gaussian components to localize celestial structures.
- **Impedance:** Terminate transmission lines to minimize reflections.
- **Propagation:** Derive propagation velocities for radio waves in coaxial cables.
- **Error Analysis:** Use linear and non-linear least-squares methods to fit models and estimate errors.

3. Schedule

This lab covers significant new territory in data analysis. Work hard to obtain good observations early. Reserve plenty of time for analysis and re-observation.

1. *Week 1:* Finish §6 and §7. Understand §4. Show your work, software, and results in class.
2. *Week 2:* Finish §8, §9, and §10. Produce candidate plots and analysis results.

3. *Week 3:* Finish re-observations. Write your formal report following the guidelines and addressing §2.

4. What Time Is It?

Relativity dictates that time depends on the reference frame. Nothing drives this home like trying to point a telescope from the surface of an orbiting, spinning sphere. Humans have invented a surfeit of time standards; we focus on those useful for astronomy.

Coordinated Universal Time (UTC) is the Civil Time in Greenwich, England. It is 8 hours ahead of Pacific Standard Time (PST). $\text{PST} = \text{UTC} - 8 \text{ hr}$; $\text{PDT} = \text{UTC} - 7 \text{ hr}$. (Warning: we switch to Pacific Daylight Time (PDT) mid-semester.) UTC measures solar time (24 hours per solar day).

Unix Standard Time measures seconds since midnight Jan 1, 1970, UTC. This is the internal clock for computers (e.g., `time.time()`). Computers use Network Time Protocol (NTP) to synchronize local quartz oscillators with atomic clocks.

Sidereal Time is “star time.” It tracks the rise and set of distant stars. The sidereal day is $1/365.24$ shorter than the solar day (about 4 minutes). This difference arises from Earth’s orbit around the Sun. Sidereal time depends on Earth’s spin, which slows due to tidal friction. (Where does the angular momentum go?) The conversion between Civil Time and LST has to be adjusted periodically by inserting *leap seconds*.

Local Sidereal Time (LST) adjusts sidereal time by longitude. A star with Right Ascension (RA) transits the local meridian when $\text{LST} = \text{RA}$.

The **Julian Day (JD)** is a sequential count of solar days since Jan 1, -4713 . It specifies date and time as a double-precision float without months or leap years. JD begins at noon, UTC.

The **Modified Julian Day (MJD)** is a shortened JD. MJD is 12 hours offset from JD to align with UTC. $\text{MJD}=0$ corresponds to 0 hr UT on 17 Nov 1858.

4.1. Useful Python Procedures for Time Conversion

Python has several built-in modules for dealing with time, including `time` and `datetime`. Astronomers, however, have historically been the best time-keepers on the planet. For astronomy-quality time routines, `astropy.time`, part of the `astropy` package, is the industry standard. We have wrapped some of them into `ugradio.timing` for your convenience. *However*, the times you use for input are only as good as the synchronization between the clock you are using (e.g. on your Raspberry Pi) and our national-standard atomic clocks, so before observing, you are going to want to use a Network Time Protocol (NTP) daemon to discipline your local clock to an external reference, and in case it isn’t clear from the name, you will have to have an active network connection.

Calculating LST requires knowing your longitude. New Campbell Hall (NCH) is at `(lat,lon) = (37.8732, -122.2573)`, Note that `lon = -122.2573 = (360 - 122.2573) = 237.7427`. This information is stored in the `ugradio.nch` module. For all procedures mentioned below, this location is default

```
local_now = ugradio.timing.local_time() # current local time as a string
ut_now = ugradio.timing.utc() # current UTC as a string
ut_now = ugradio.timing.unix_time() # seconds since 1 January 1970
julian_now = ugradio.timing.julian_date() # current julian day (which \
```

contains the current time, too--it's not just an integer \ number.

```
lst_now = ugradio.timing.lst() # current LST at NCH
lst_julian = ugradio.timing.lst(jd) # LST for the specified Julian day
ut_julian = ugradio.timing.unix_time(jd) # seconds since 1/1/1970 for given JD
julian_ut = ugradio.timing.julian_date(ut) # julian day for given unix time
```

5. Handouts

Our website has topical handouts that may be of use to you:

1. The complete story of producing calibrated spectral line shapes and intensities: *CALIBRATING THE INTENSITY AND SHAPE OF SPECTRAL LINES*. This handout is optional because the steps discussed in the lab instructions are sufficient, but if they are unclear or you want to know more, this writeup has it all.
2. How does the power we receive with a telescope relate to a source being observed? It's all about specific intensity, usually denoted I : "SPECIFIC INTENSITY: THE FOUNT OF ALL KNOWLEDGE!". You need to understand these topics to interpret your measurements.
3. Astronomical coordinate transformations with Rotation Matrices: "SPHERICAL/ASTRONOMICAL COORDINATE TRANSFORMATION". converting among Az/El, Ha/Dec, Ra/Dec, L/B. You need to know coordinates for all the remaining labs.

6. Your First 21-cm Measurement (Group Lab Activity, Week 1)

Use the techniques we explored in Lab 1 to observe the 21-cm line of neutral hydrogen (colloquially, **HI**, pronounced "H-one"), measuring its line shape, velocity, and intensity using the big horn on the New Campbell Hall roof. Most of the measured power comes from noise in our own electronics, not the HI line. It's often called noise, and we need to calibrate out this instrumental contribution and the responses of our amplifiers and filters to obtain the correct line shape and intensity.

Measure the 21-cm line twice. The first time, the goal is to master the technical aspects of the system and procedures, so instead of worrying about where to point the horn, just use whatever sky position happens to be overhead. The second time (§8), manually point the horn to a designated position and make a calibrated profile to compare with a well-established standard profile measurement.

6.1. The Receiving System

After analog amplification and filtering, we need to use our Raspberry Pi / SDR system with its built-in digital downconverter. Having already mastered the intricacies of DSB/SSB mixers and filters in the first lab, we are ready to jump to a higher level of abstraction with dedicated hardware for down conversion.

6.1.1. Digital Down-Conversion

The NeSDR SMART modules we have in the lab are based on two chips: the R820T2, which contains a voltage-controlled oscillator and mixer (i.e. a DSB with a tuneable LO), and the RTL2832U, which contains a 28.8 Msps ADC, two multipliers (mixers) fed with digital sine/cosine waves, and on-board FIR (convolution) filters. In other words, the RTL2832U contains an all-digital SSB mixer/filter down-converter.

Using the `rtlsdr` driver that has already been installed on your RPi and the `pyrtlsdr` bindings for Python control, you can set the LO frequency of the R820T2 DSB mixer. We suggest setting your LO off to the side of the 1420.405 MHz line, since the 0-frequency bin in your spectrum often has a systematic spike in it. How far you can tune your LO away from the rest frequency depends on the sample rate (read: Nyquist cutoff) you choose and the width of the analog filters upstream.

As you saw in Lab 1, you can also adjust the sample rate at the output of the RTL2832U, which controls how many of the 28.8 Msps samples get through. This is called down-sampling, and it is necessary to reduce the data bandwidth enough that a USB interface can carry it over to the RPi. If you set the sample rate appropriately (and use the fast USB-3.0 ports) it is possible to stream data through your RPi without losing any samples.

Ultimately, you are going to need to acquire many samples in order to reduce the noise in an integrated spectrum enough to see the HI line. How long that takes depends on the efficiency of your code, so pay attention to block size, sample size, and how often you save data to disk. Avoid re-initializing software interfaces where possible, but also mitigate the risk of losing your entire observation if a connector gets jostled or an unexpected Python exception gets thrown.

6.2. Your Measurement

Before taking data, do basic system checks: (1) use oscilloscopes and captured voltage waveforms to set signal levels appropriately, and (2) experimentally verify that you are looking at the frequencies you intend to. In order to test that you are seeing data through an active (and powered!) signal chain, you can use the tone-injection system to broadcast a known RF signal into the front of the horn. Make sure you are seeing the actual signal by moving it around in frequency, turning it on/off, or changing its amplitude to make sure the “birdie” you put in is where you expect in your spectrum. You should notice that your birdie amplitude can be quite low; we have a fairly sensitive receiver system.

Once you are convinced your signal is getting through, set the system up as you will be using it to observe. Point the horn to zenith to reduce interference and thermal noise. Take some data. How fast must you sample?

Look at the range of sample values using, e.g., `np.histogram`. As in Lab 1, check that you are not heavily quantized or clipping. Use analog attenuators and/or digital gain in `sdr.capture_data` to get the levels right. The histogram shape should look like a familiar function. Does it?

As a final note, remember that in inclement weather, the horn is a gigantic rain bucket. You won’t see any astronomical signal unless you dump it!

6.2.1. Planning Observations

Having determined that the system basics work, it’s time to do astronomy! For this first measurement the main goal is to master the technical aspects, so point the horn to whatever piece of sky happens to be overhead. The line will be strongest—strong enough to see visually—in the range $\text{LST} = 19 - 6$ hr.

It is convenient to use temperature units for the power that we measure. The power we measure is called the **system temperature** T_{sys} , and it is a function of frequency. It contains two components: the **continuum**, which is devoid of spectral features and changes slowly with frequency; and the **line**, which in this case is from hydrogen, and changes relatively rapidly with frequency—hence our desire to obtain the line shape.

The continuum system temperature has two contributions: the dominant contribution from our electronics, which we call the **receiver temperature** T_{rx} ; and a contribution entering our antenna from the sky, the **sky temperature** T_{sky} . (This is also sometimes called the **antenna temperature**). Thus $T_{\text{sys}} = T_{\text{rx}} + T_{\text{sky}}$, and above 300 MHz, usually $T_{\text{rx}} \gg T_{\text{sky}}$.

Our horn is equipped with an ambient-temperature first amplifier so $T_{\text{rx}} \sim 300$ K. (For Lab 4, our Leuschner telescope is much better, with $T_{\text{rx}} \sim 50$ K). The sky temperature comes from the Cosmic Microwave Background ($T_{\text{CMB}} \approx 2.7$ K); from interstellar/intergalactic space, with brightness temperature T_{IGM} (usually no more than a few K in the L-band continuum and up to about 100 K in the HI line); and the Earth’s atmosphere with brightness temperature T_{atm} , perhaps a few K at the HI line frequency. So off of the HI line we have $T_{\text{sky}} \sim 10$ K, while on the HI line, in the Galactic plane where it is strongest, we can have $T_{\text{sky}} \sim 100$ K.

6.2.2. Two Frequency-Switched Line Measurements

The measured power spectrum shape is dominated by filters acting on the system temperature. To see the line, which is weak, we need to correct for these filter shapes, which can be characterized by taking data away from the spectral line. Take two sets of data:

1. a long integration to measure the HI line profile (the on-line spectrum s_{on}), and
2. a short integration with the line not present (the off-line spectrum s_{off}) to calibrate the absolute *intensity*.

You can obtain the on-line spectrum by centering the line and making a measurement; then changing the first LO frequency so that the line shifts either completely outside the band (this is called **frequency switching**), or partway over but is still in the band (in-band frequency switching). The latter is better because you are always looking at the line; you end up with more measurements and better signal/noise.

To accomplish this, take a spectrum with the line in the upper half of the downconverted (baseband) spectrum, and another with it centered roughly in the lower half. Use the first as the on-line and the second as the off-line spectrum for the upper half. Similarly, for the lower half, use the second as the on-line and the first for the off-line. (The HI line frequency is 1420.4058 MHz).

The line is weak, so take lots of spectra. Recall that `sdr.capture_data` obtains a specified number of blocks (`nblocks`), each of which has `nsamples` datapoints. You might be tempted to do a deep observation by setting `nblocks` to a large number. That may work, or you may run into memory or stability issues, and if Python throws an error, you could lose all of your data. Maybe you would prefer to spread your eggs over more baskets. Either way, I strongly recommend writing exception handling into your code.

6.2.3. Two Frequency-Switched Reference Measurements

Intensity calibration requires a second set of (short) measurements with your telescope aimed toward a source of known temperature. Easiest is probably to take one with the horn looking at a known blackbody and one looking at the cold sky. What’s a convenient blackbody? You and your friends! Take one short measurement with the horn pointing straight up at the cold sky and the other with as many people as you can fit standing in front of it to fill the aperture. Call these spectra s_{cold} and s_{cal} , respectively.

7. Analysis (Individually at Home, Week 1)

7.1. Take a Suitable Average/Median

Consider, first, the s_{on} and s_{off} spectra, from which you can find the line shape. There are many individual spectra, which you need to combine these to make a single spectrum for each measurement. You can do this by averaging the power spectra (use `numpy`’s `mean` function) or by taking the medians (use `numpy`’s `median` function). The former gives a less noisy result, but the latter handles time-variable interference better; use both and compare the results.

Even after combining many spectra, the resulting spectrum will look noisy. You can reduce the noise by averaging over channels, but degrades the spectral resolution. Find a compromise on how many channels to smooth over, assuming that the HI line is never narrower than about 1 km/s. Again, try both `np.mean`) and `np.median`) to see what happens.

In the smoothed s_{on} spectrum, you might not be able to see the HI line because the instrumental bandpass dominates the spectrum shape. The instrumental bandpass is determined mainly by the SDR’s FIR filter, which should fall smoothly away from zero frequency. Does it reach zero at the band edge? If not, should you worry about aliasing? What does that tell you about where to place the HI line?

7.2. Get the Line Shape

You can remove the instrumental bandpass to get the shape of the line r_{line} (but not the intensity), by taking the ratio

$$r_{\text{line}} = \frac{s_{\text{on}}}{s_{\text{off}}} . \quad (1)$$

This is the first factor (i.e., the shape) in equation (13) in the spectral-line handout. Note that the power units of s (and any calibration constants) divide out in r , which is unitless.

7.3. Get the Line Intensity

To get the line intensity in temperature units, we need to use a calibration noise source to determine the **gain**—the second factor in the handout’s equation (13). We obtain the gain, G , by using a known difference in system temperature between the the calibration ($T_{\text{sys,cal}}$) and the cold sky ($T_{\text{sys,cold}}$), and seeing how much that known temperature difference changes the measured values in your spectra, as per equation (15) in the handout:

$$G = \frac{T_{\text{sys,cal}} - T_{\text{sys,cold}}}{\sum (s_{\text{cal}} - s_{\text{cold}})} \sum s_{\text{cold}} \quad (2)$$

Here, a sum is over all channels in a power spectrum. This equation converts measured units, which are proportional to power, into physically meaningful units, i.e. Kelvin. Assume $T_{\text{sys,cal}} = 300$ K, because that’s the thermal power you injected by standing in front of the horn. Since $T_{\text{sys,cold}} \ll T_{\text{sys,cal}}$, you can, to a reasonable approximation, neglect it (which is what we did in the handout). Then the final, intensity-calibrated spectrum is equation (16) in the handout, namely

$$T_{\text{line}} = r_{\text{line}} \times G \quad (3)$$

Note how G carries the units of temperature.

7.4. Plotting Intensity vs. Frequency—and Velocity

Plot your calibrated spectrum versus RF frequency. Next, plot it versus the Doppler velocity¹. Remember that, by astronomical convention, we have placed ourselves in the center of the universe, so positive velocity means motion away from us:

$$\frac{v}{c} = \frac{\Delta\lambda}{\lambda_0} \approx -\frac{\Delta\nu}{\nu_0}, \quad (4)$$

where c is the speed of light and $\Delta\nu$ is the frequency offset from the line frequency ν_0 .

7.5. Finally, Choose a Reference Frame

The velocity you just derived depends on time of year, and your position on Earth. To make a reproduceable measurement, we need to correct the observed velocity for the orbital velocity of the Earth and the Earth’s spin. When observing the Galaxy, it is customary to express velocities with respect to the **Local Standard of Rest (LSR)**.

Calculate the Doppler correction using `ugradio.doppler.get_projected_velocity`. Correct the velocities and compare the spectrum for the observing frame and the LSR frame, which is approximately the frame that would rotate around the Galaxy in a circular orbit. Correcting to the LSR involves many components, including primarily: the rotation of the observatory around the center of the Earth, the orbit of the Earth around the barycenter of the solar system, and the peculiar velocity of our Sun with respect to other stars in the neighborhood. There are higher-order corrections (which are in the `barycorrpy` package, see (Wright & Eastman, 2014)), including 1) special relativistic treatment of velocity, 2) general relativistic effects from the influence of the gravitational fields of all bodies in the solar system, and 3) the proper motion of the target source.

Notes on `ugradio.doppler`:

1. You need the celestial coordinates of your target (ra, dec). How to find these for the horn pointing straight up?
2. You need the Julian day of your observation; see §4.
3. You need the observatory coordinates (north latitude and west longitude) in degrees; you could enter them with the pair of optional input parameters (`obs_lat`, `obs_lon`), but you don’t have to because the default values are Campbell Hall’s values (which are `lat`= 37.873199 `lon`= −122.2573 degrees).

8. Your Second 21-cm Line Measurement (Group Lab Activity, Week 2)

Obtain a fully-calibrated spectrum for the horn pointing at Galactic coordinates $(l, b) = (120^\circ, 0^\circ)$. How do you know where to point the horn? You should use rotation matrices (which you can check against `astropy`. See the handout “Spherical Coordinate Transformation” (§5). Plot your spectrum versus both the observed and the LSR velocity.

Your spectrum might look something like that in the top panel of Figure 1. There is a vertical offset, a curvy baseline, and one or more spectral peaks. The curvy baseline comes from our instrumental response, and can be removed with (in order of rigor) a low-order polynomial fit, an interpolated Fourier fit, or by fitting a covariant eigenmode pseudoinverse to the off-line channels (the approximate frequency ranges $-1.2 \rightarrow -0.6$ and $0.6 \rightarrow 1.2$ MHz).

¹Astronomers usually express velocities in km s^{-1} .

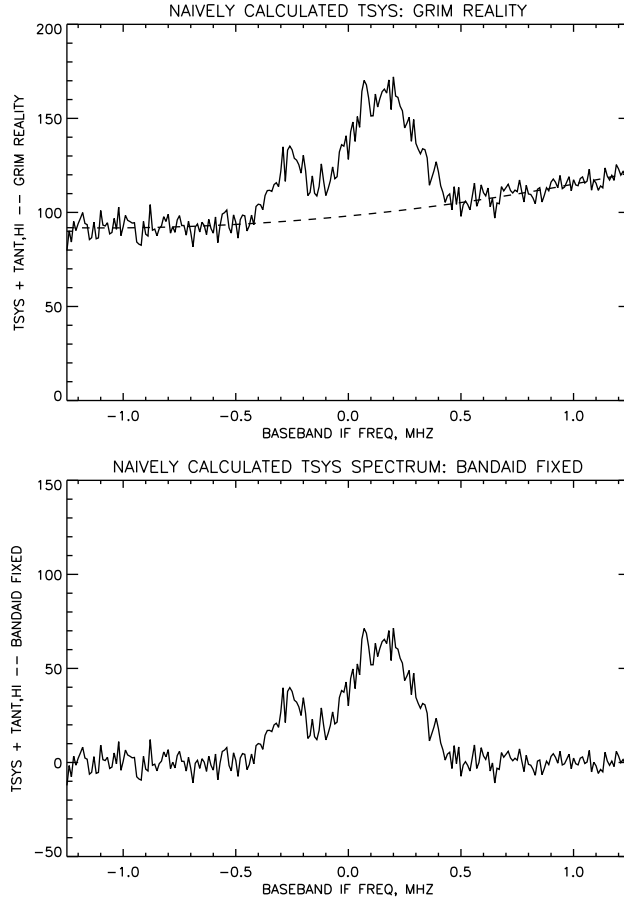


Fig. 1.— A typical raw spectrum with a curvy baseline and multiple velocity components.

The dashed line is a second-order polynomial fit to the off-line channels and the bottom panel has this fit subtracted off. Here, the multiple peaks come from HI clouds having different velocities. The peaks are usually represented as Gaussians with appropriate central intensities, velocities, and widths. Fitting Gaussians requires specifying initial guesses for the component parameters. These guesses need not be particularly accurate. For example, in the bottom panel of Figure 1, initial guesses for the two heights, centers, and widths (FWHM) could be $[20, 50]$, $[-0.3, 0.2]$ MHz, and $[0.01, 0.03]$ MHz.

After fitting, check the quality of the result. If it is struggling, that is usually because you need one or more additional components. It is common to require a low-level, broad component that produces no obvious peak; here, the initial guesses might be $[20, 50, 10]$, $[-0.3, 0.2, 0.]$ MHz, and $[0.01, 0.03, 0.07]$ MHz.

These procedures may help you for these least-squares fits, but also feel free to roll your own data-reduction techniques. Some people in past years have applied Markov Chain Monte Carlo fitters to generate posterior parameter probabilities.

1. For the polynomial fit, use `np.polyfit`

2. For fitting multiple Gaussians, use our home-grown `gaussfit`; for evaluating the Gaussians from the parameters produced by `gaussval`, use `gaussval`.

In class, we will compare the results from all groups. How reproducible is our science?

9. Measuring the Speed of Light in Cables (Group Lab Activity, Week 2)

For moving power or electronic information from one place to another, *transmission lines* (cables) are indispensable. It sounds easy—just connect two things with a wire. But does this really work? We’ll explore cables and reflections by measuring the Voltage Standing Wave Ratio (VSWR), which results from interference of incident and reflected waves.

The experimental work and measurements described below should be done by groups. The analysis in §10 should be done by individuals. The analysis is nontrivial, so don’t delay with the measurements!

9.1. Estimating Receiver Noise and Gain

Our receiver system consists of a horn antenna, three amplifiers, a long length of 50Ω coaxial cable down from the roof, and then whatever additional amplifiers and filters you see in the rack. In order to obtain a trustworthy calibration, we must account for the signal loss, gain, and noise contributions of these components.

In §6.2.3 we performed an end-to-end calibration by pointing the antenna into a known load (you, a ~ 300 K blackbody). Our goal here is to derive an independent cross-check on that calibration by building up a model of our signal path. To do this, we need to think about the individual components in our system, as well as the transmission lines that separate them.

Start at the end (your SDR) and work backwards.

9.2. Characterizing the Gain Scale of the SDR

As described in class, coaxial cables propagate waves by providing a uniform (50Ω , ideally frequency-independent) impedance along their length, so that each section of cable receives the outgoing wave from its upstream neighbor with the same impedance with which it was transmitted. This prevents reflections from occurring along the length of the cable, but they can still arise if you do not terminate the cable with the same characteristic impedance.

The end of the line in our receiving system is our SDR module, and we need to check its input impedance to relate the voltages we measure to signal power. This allows us to calibrate our digitized data.

Inject a signal of known power into the SDR. If you are worried about cable losses (which are particularly important on long cables), you can use an oscilloscope to measure the signal that your SDR sees, but beware. The oscilloscope uses $1\text{ M}\Omega$ termination, effectively acting like you’ve left the end unconnected. To terminate properly, you will need to use a T junction to put a 50Ω terminator in parallel with the oscilloscope.

Next, check the SDR termination to ensure you are making an apples-to-apples voltage comparison with the oscilloscope. The easiest way to measure if a cable is properly terminated is to send a signal in one end and measure if a reflection of that signal comes back out the same end. Connect the oscilloscope and a function generator (outputting a square wave) to one end of a long cable. For the moment, leave the far end of the cable unconnected (“floating”).

Zooming in on time with your oscilloscope, you should see a reflection of the square wave

interfering with the input wave. Notice the time lag between the input and reflected waves? That lag corresponds to the time it takes to traverse the cable twice. We will use this later to calculate the speed of light in a cable. Now try terminating the far end of the cable. Did the interference pattern go away? It should have, because that’s the job of termination: to prevent reflections.

Repeat this experiment, but with your SDR on the far end of the cable. Does it properly terminate the cable, or should you add your own 50Ω terminator? Does it change if you plug the SDR in to your RPi? Be aware that impedance can change versus frequency, so you may find that low-frequency signals exhibit a different termination than high frequencies. What frequency should you use? How would that affect the waveform you see on the oscilloscope?

Using everything you know about termination, calculate (with error bars!) the voltage scale of your SDR, the speed of light on a cable, and the signal loss per meter through a coaxial BNC cable at 1420 MHz. In doing your SDR calibration, be sure to keep track of the LO and RF frequency settings you use, because we also need to calibrate the spectral response of your SDR relative to this measurement.

9.2.1. A Note on FIR Filters

An FIR filter works as an anti-aliasing filter by convolving 28.8 Msps digitized data in the RTL2832U before it is decimated (reduced in sample rate by throwing out samples) to achieve the sample rate you specify. The default coefficients are:

$$\begin{aligned} &-54, -36, -41, -40, -32, -14, 14, 53, 101, 156, 215, 273, 327, 372, 404, 421, \\ &421, 404, 372, 327, 273, 215, 156, 101, 53, 14, -14, -32, -40, -41, -36, -54 \end{aligned}$$

(Note the mirror symmetry.) You can change these if you choose. The 16 numbers you enter are mirrored to generate the full FIR filter.

The Convolution Theorem implies that the expected (voltage-spectrum) frequency response is the Fourier transform of these coefficients. That means you can use `ugradio.dft` to sample the filter response at the frequencies relevant for the Fourier Transforms you will be taking.

With the signal you calibrated and the FIR filter shape you calculated (or the square of it, for power spectra), you can partially calibrate an input signal to your SDR module. However, there appears to be another summing filter in the RTL2832U that is not well documented. I estimate its coefficients to be something like:

$$-1/8, -1/4, -3/4, -1/2, -1, 8, -1, -1/2, -3/4, -1/4, -1/8$$

Again, you would use the square of the DFT of these coefficients to correct the remaining ripple in your power spectrum bandpass shape, after applying the previous filter correction.

You’ll notice my coefficients aren’t perfect. Can you do better? Check how well your coefficients flatten out a white-noise signal.

9.3. Characterizing Amplifiers, Filters, and Cable Loss

Following the above steps, we calibrated the input to the SDR, but that isn’t enough to calibrate our telescope. The sky signal passes through amplifiers, filters, and a long cable on the way to our SDR. We need to calibrate those as well.

Up at the horn, there are 3 or 4 amplifiers (some in the box at the base, and one up near the back of the horn). Find their part numbers and look up the gain (in dB) on their respective datasheets online.

Next, you need to estimate how much signal is lost in the cable running down from the roof to the 5th floor lab. Fortunately, you have already characterized the loss in BNC cables per length. All you need to do is figure out how long the cable is, which you can do using the same square-wave reflectometry you used in §9.2.

Finally, characterize the gain/loss through the bandpass filter and homebrew amplifiers in the lab. This can be done by injecting a known signal amplitude and measuring the output with your newly calibrated SDR module.

9.4. Antenna Efficiency

We’ve characterized the gain in almost every element of our system, except for one: the antenna. Antennas can be thought of as an impedance-matching circuit that connects the transmission line of free space (with a characteristic impedance of 377Ω) to a coaxial transmission line, usually with the help of a balun matching circuit.

In practice, the impedance match of an antenna is imperfect and highly frequency-dependent. As a result, not all of an incoming sky signal makes its way into our receiver; a substantial fraction may reflect and scatter back out. To account for this, we need an **aperture efficiency** coefficient that, given the physical area of our antenna’s aperture, scales how much flux collected over that area actually enters our electronics. Typical well-matched antennas have efficiencies of 0.6 to 0.7, but poorly matched antennas can be much lower than this.

Let’s assume we have a well-matched antenna, but be prepared to revisit this assumption using the results of your end-to-end blackbody load calibration. If you are clever (and careful), you may also be able to use the tone injection system to further constrain the aperture efficiency.

If you wish to disconnect or rewire anything at the antenna (to, for example cross-check the amplifier gains from the datasheet), you may, but make sure you leave the system working for the next group.

10. Statistical Error Analysis (Individually At Home, Week 2)

It’s time to perform a rigorous, end-to-end calibration of our receiver system, with the goal of reporting (with correct error bars), the amplitude (in brightness temperature), Doppler shift (in km/s), and Doppler width (in km/s) of the spiral arms we observed.

Using your blackbody load calibration and your receiver characterization to quantify the aperture efficiency of our horn antenna. Estimate the error in your calibration multiple ways and check for consistency between them. Estimate the error associated with your removal of the baseline system temperature. All of these systematic errors contribute to our uncertainty.

We also have random (thermal) noise contributing to our uncertainty. Quantify the noise levels in each individual spectral measurement, per frequency channel. Ideally, your resultant Gaussian fits should go through your data points to within the thermal noise. Calculate the reduced chi-square of your fit. Is it close to one? Can you use χ_r to argue whether to use fewer or more Gaussian components in your model?

Finally, estimate the error in your parameter fits using any of the tools we presented in class. You may do an MCMC fit, per the notebook example, or quantify a 2-sigma chi-square interval, or use the covariance matrices of a parameter-fitting package. Crucially, include systematic errors on top of the statistical errors, and make sure, at the end of the day, that you believe your result. Where possible, test your ability to recover true answers in numerical simulations to validate your

process.